Exercise 1 Show that the Voronoi classifier can be written as

$f(x) = \frac{1}{2}\left(\| x - C_- \|^2 - \| x - C_+ \|^2\right)$

Defining $C = (1/2)(C_+ + C_-)$ and expanding this equation gives

$f = \frac{1}{2}(\langle x - C_-, x - C_-\rangle - \langle x - C_+, x - C_+\rangle) =$
$\frac{1}{2}\left(\langle x, x\rangle^2 - 2\langle x, C_-\rangle + \langle C_-, C_-\rangle - \langle x, x\rangle^2 + 2\langle x, C_+\rangle - \langle C_+, C_+\rangle\right) =$
$\frac{1}{2}(2\langle x, C_+\rangle - 2\langle x, C_-\rangle + \langle C_-, C_-\rangle - \langle C_+, C_+\rangle) =$
$\langle x, C_+\rangle - \langle x, C_-\rangle + \frac{1}{2}\langle C_-, C_-\rangle - \frac{1}{2}\langle C_+, C_+\rangle =$
$\langle x, C_+ - C_-\rangle + \frac{1}{2}\langle C_-, C_-\rangle - \frac{1}{2}\langle C_+, C_+\rangle =$
$\langle x - C + C, C_+ - C_-\rangle + \frac{1}{2}\langle C_-, C_-\rangle - \frac{1}{2}\langle C_+, C_+\rangle =$
$\langle x - C, C_+ - C_-\rangle + \langle C, C_+ - C_-\rangle + \frac{1}{2}\langle C_-, C_-\rangle - \frac{1}{2}\langle C_+, C_+\rangle$

Now the last two terms cancell since
$\langle C, C_+ - C_-\rangle + \frac{1}{2}\langle C_-, C_-\rangle - \frac{1}{2}\langle C_+, C_+\rangle =$
$\frac{1}{2}\langle C_+, C_+ - C_-\rangle + \frac{1}{2}\langle C_-, C_+ - C_-\rangle + \frac{1}{2}\langle C_-, C_-\rangle - \frac{1}{2}\langle C_+, C_+\rangle =$
$\frac{1}{2}\langle C_+, -C_-\rangle + \frac{1}{2}\langle C_+, C_-\rangle = 0$

Excercise2 For the Voronoi classifier example, the kernelized f is given by

$f(x) = \sum_i \alpha_i k(x, x_i)$, with

$$\alpha_i = \begin{cases} -(y_i/n_n), \text{if } x_i \text{ was misclassified as postive in training} \\ (y_i/n_p), \text{if } x_i \text{ was misclassified as negative in training} \\ 0, otherwise \end{cases} \tag{1}$$

$$b = \frac{1}{2}\left(\frac{1}{n_n^2}\sum_{i|y_i<0}\sum_{j|y_j<0} k(x_i, x_j) - \frac{1}{n_p^2}\sum_{i|y_i>0}\sum_{j|y_j>0} k(x_i, x_j)\right)$$

Exercise 3 Show that $f_\phi(x) = \frac{1}{2}(\| \phi_x - \phi_n \|_H - \| \phi_x - \phi_p \|_H)$

Following the details from Exercise 1

If we take $\phi_{x*} = \phi_{C+-C-}$ and $b = \frac{1}{2}\left(K(C_+, C_+)^2 - K(C_-, C_-)^2\right)$, then the sums over x_i are taken before the mapping $\phi : \chi \to H$ and f takes the form

$$f_\phi(x) = \langle \phi_x, \phi_{x*}\rangle + b.$$

A property that $\phi(x)$ should have is $\phi(x) : x \to h(x - C)$ so that h does shifting by C in calculating the inner product. In this way we model the defintion of f from exercise 1.

Exercise 4
$\chi = \{(e2, e2), (e2, e3), (e3, e3), (e3, e4), (e4, e4)\}$.
Set each element of χ to be the vector x_i, for $i \in \{1, \ldots, 5\}$.

The kernel matrix, K given by

$$K = \begin{bmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 0 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Its eigenvalues are λ=(4.3027756,2.6180340,2.0000000,0.6972244,0.3819660).

Prove that K is a positive definite kernel.

K is positive definite since all its eigenvalues are positive.

Exercise 5
Find a Hilbert Space H and a mapping ϕ such that $K(x, y) = \langle \phi_x, \phi_y \rangle$.
Consider $H = \mathbb{R}^6$ and set

$$\begin{aligned}
\phi(x_1) &= (1, 1, 0, 0, 0, 0) \\
\phi(x_2) &= (0, 1, 1, 0, 0, 0) \\
\phi(x_3) &= (0, 0, 1, 1, 0, 0) \\
\phi(x_4) &= (0, 0, 0, 1, 1, 0) \\
\phi(x_5) &= (0, 0, 0, 0, 1, 1)
\end{aligned}$$

Then for any x and $y \in \chi$, every element of K can be represented as
$k(x, y) = \langle \phi(x), \phi(y) \rangle$.

Exercise 6
Show that $k(x, y) = x^\top y$ is pos. definite.

$$v^\top K v = \sum_{ij}^{n} v_i v_j k_(x_i, x_j) = \sum_{ij} v_i v_j \left(x_i^\top x_j \right)$$

$$= \sum_i \sum_j v_i v_j x_i^\top x_j = \left(\sum_i v_i x_i^\top \right) \left(\sum_j v_j x_j \right)$$

$$= \left(\sum_i v_i x_i \right)^\top \left(\sum_j v_j x_j \right) \geqslant 0$$

If set $w = \sum v_i x_i$ then last term is $\|w\|$ which is greater or equall to zero.

Assume $x \in \mathbb{R}^d$ and set $k(x,y) = x^\top y$.

The construction of the associated Hilbert space will follow discussion in class.

$f(x) = \sum_i^n k(x, x_i) = \sum_i \alpha_i x^\top x_i = \sum_i \alpha_i x_i^\top x =$

$\left(\sum_i \alpha_i x_i\right)^\top x = v^\top x$

Guess: H $= \{f_v | f_v(x) = v^\top x, v \in \mathbb{R}^d\}$.

Check to see if this construction has the correct properties:

0. Since by construction $v \in \mathbb{R}^d$, the Hilbert Space is defined
on vectors of finite dimension and is hence is complete.

1. Now we check to see if $k(x,y) = x^\top y$ implies an f of the form f_v.

Let $x \in \chi$. Since $x \longrightarrow x^\top y = y^\top x = f(x)$ for $y \in \mathbb{R}^d$,
$k(x,y)$ will produce f of the form $f_v \in H$.

2. Check that the k is a reproducing kernel. I.e $\langle f_v, k(\cdot, y)\rangle = f_v(y)$ Since

$f_v(y) = v^\top y$,
$\langle f_v, k(\cdot, y)\rangle = \langle f_v, f_y\rangle = v^\top y = f_v(y)$

So k is reproducing.

Excersize 7

The quadratic kernel given by $\left(x^T y\right)^2$ is pos. definite.

Proof: Take any vector $v \in \mathbb{R}^n$. Then the quadratic form is given by
$v^\top K v = \sum_{ij} \alpha_i \beta_j K_{ij} = \sum_{ij} \alpha_i \beta_j \langle \phi(x_i), \phi(x_j)\rangle_H$. Since for any pair of points with components given by $x = (x_1, x_2)$ and $x' = (x_1', x_2')$,
the kernel reduces to $(x_1 x_1' +_2 x_2')^2$ and each element is symtetric with respect to the primed coordinates. This means K is as well. Then by construction K can be decomposed as $K = A^\top A$ and the quadratic form becomes

$v^\top K v = v^\top A A^\top v = \left(v^\top A\right)\left(A^\top v\right) = \left(A^\top v\right)^\top A^\top v = \| Av \|^2 \geq 0$